

ACTIVITY 12 — FACILITATOR GUIDE**Network Diagram, PERT and the Critical Path**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO2	3 - Plan the Project

Purpose of this activity

By the end of this activity learners will be able to derive an activity list from the WBS work packages, converting deliverables into the activities that produce them.

What learners produce

activity list, network diagram, critical path — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- The Project Management Templates pack (see the Activities folder)
- Flipchart or A3 sheet and markers
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Derive the activity list from the WBS. Lab 11 produced deliverables. A schedule needs activities - the verbs that produce those nouns. This is the conversion the WBS deliberately did not do.	Watch for groups listing only the obvious. Ask what they are deliberately leaving out, and why.
2	Understand the dependencies you have just declared	If a group finishes early, ask what changes at twice the scale or half the budget.
3	Draw the network diagram. Create artifacts/12-network-diagram.md. This is an activity-on-node diagram: activities are boxes, arrows are dependencies.	Have one group present a single row rather than the whole artifact — depth over coverage.
4	Compute the forward and backward pass. Do this by hand before reading the table. The method:	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.

5	Apply three-point estimating and compute confidence ranges. Single-point estimates hide uncertainty. Three-point estimating exposes it.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
6	Compress the schedule. Crash cost table. All four activities below are on the critical path.	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Apply the Lab 03 lesson: compliance before UAT. The Lab 03 lessons-learned register recorded that on the previous release, compliance review was run concurrently with user acceptance testing to save time, and the findings forced rework of screens that had already been accepted by users. The lesson: compliance review must complete before UAT begins.	Watch for groups listing only the obvious. Ask what they are deliberately leaving out, and why.
8	Build the schedule baseline. Create artifacts/12-schedule-baseline.md.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
9	Answer the exam-style scenarios	Ask what the number would have to be before the decision flips. That is the exam's favourite question.

Suggested timing

Minutes	Phase
9 min	Brief the scenario, confirm the inputs and assign roles
33 min	Work the first half of the steps individually, then pool as a group
34 min	Complete the remaining steps and finalise the deliverable
14 min	Group share-back, facilitator debrief and exam-style linkage

Total: 90 minutes

Discussion & decision prompts

1. Derive an activity list from the WBS work packages, converting deliverables into the activities that produce them — how did your group decide, and what would change that decision?
2. Sequence activities using the four dependency types and distinguish mandatory from discretionary logic — how did your group decide, and what would change that decision?
3. Compute a forward pass and a backward pass by hand and identify the critical path with total and free float — how did your group decide, and what would change that decision?
4. Your critical path is A-B-D-G-J-L-M-N-O-P at 123 days, and every activity on it has zero total float — can every member of your group defend it?
5. Where would network diagram, pert and the critical path have saved you time or money on a real project you have worked on?

What good looks like

- Your critical path is A-B-D-G-J-L-M-N-O-P at 123 days, and every activity on it has zero total float.
- Your forward pass took the MAXIMUM predecessor EF at activity J and your backward pass took the MINIMUM successor LS at activity B. Getting either backwards produces a plausible-looking but wrong schedule.
- Total float equals LS minus ES and also equals LF minus EF for every activity. If the two differ on any row, that row is wrong.
- You identified that C, F and I have 20 days of total float but zero free float, and can explain why.

- You computed project standard deviation from the critical path variances only, and you added variances rather than standard deviations.
- Your 95.45% range is roughly 113.5 to 132.5 days, and you stated it as a probability sentence rather than a number.
- Your crash selection is N plus G at SGD 12,100, chosen by cost per day saved rather than by total days available.
- You stated that crashing activity E would shorten the project by zero days, and why.